Beixi (Chloe) Du

347 807 9797 | bd469@cornell.edu | bd469.github.io

RESEARCH INTERESTS

Learning and decision-making under uncertainty, including learning from demonstrations, belief-state estimation and control, and visual reasoning and evaluation.

EDUCATION

Cornell University, College of Engineering Aug 2025 – Dec 2026

M.Eng. in Financial Engineering (expected) | GPA: 3.951 Ithaca, NY

- Selected coursework: Applied Time Series; Stochastic Calculus; Optimization; Monte Carlo Simulation; Operations Research Tools for Financial Engineering.

McGill University Sep 2021 – May 2025

B.A. in Mathematics | GPA: 3.96/4.00 Montreal, Canada

Minors in Computer Science and Economics

- Selected coursework: Applied Machine Learning; Real Analysis; Numerical Analysis; Ordinary Differential Equations; Algorithms and Data Structures; Stochastic Processes.

PUBLICATIONS AND MANUSCRIPTS

** Equal contribution*

1. The Prestige: Benchmarking Cognitive Visual Reasoning using Magic Tricks
Shuo Wen, **Beixi Du**[*], Edwin Meriaux[*], Junming Shi[*], Chloe Si[*], David Meger, Doina Precup, Gregory Dudek
NeurIPS 2026, Evaluations & Datasets Track.

2. Dual Control of Linear Systems from Bilinear Observations with Belief Space Model Predictive Control
Daniel Cao[*], **Beixi Du**[*], Andrew Lowitt[*], Sunmook Choi[*], Sarah Dean, Yahya Sattar
arXiv preprint arXiv:2604.24663.

3. Retriever-Free Retrieval: Navigating Documents with Decision Models
Chao Wan[*], Paul Jünger[*], **Beixi Du**, Bodong Liu, Navaneetha Krishnan, Kilian Q. Weinberger
arXiv preprint arXiv:2610.35868, 2026.

RESEARCH EXPERIENCE

UCLA Artificial Intelligence and Visual Computing (AIVC) Lab Oct 2026 – Present

University of California, Los Angeles, Department of Mathematics
Supervisors: Prof. Chenfanfu Jiang and Yumeng He

Cornell University, Department of Computer Science Sep 2025 – Nov 2026

Advisor: Prof. Kilian Q. Weinberger

Retriever-Free Retrieval: Navigating Documents with Decision Models Sep 2026 – Nov 2026

- Extended DecisionRAG from MultiDoc2Dial to ConditionalQA and MTRAG-IBM Cloud; built corpus pipelines, adapted hierarchical retrieval stages, and conducted design-set tuning and held-out evaluations.
- Designed an evaluation protocol with result-file integrity checks, dialogue-level paired bootstrap confidence intervals, and standardized per-query latency accounting.
- Implemented lexical, dense, learned-sparse, cross-encoder, and LLM reranking baselines; compared against published task-trained systems and alternative relevance judgments.

LLM Agent Policy Recovery from Noisy Human Demonstrations Sep 2025 – Oct 2026

- Recovered reusable decision rules from 74 noisy human-demonstration trajectories using graph-based extraction and cross-trajectory statistical gap detection; ablated graph refinement, delta-batch updates, and failure-contrast sharpening.
- Raised pass@1 from 67.5% (ACE low-quality baseline) to 90.0%, approaching the 95.0% high-quality-demonstrator reference; pass@3 consistency increased from 42.5% to 55.0%.

Belief-Space Learning and Control Sep 2025 – Oct 2026

Cornell University, Department of Computer Science
Advisor: Prof. Sarah Dean

Belief-Space Control for Bilinear Observation Systems Sep 2025 – Apr 2026

- Studied dual control with action-dependent observation matrices, where an action affects both the system state and the quality of future observations.
- Implemented separation-based control, model predictive control (MPC), and belief-space MPC in PyTorch with L-BFGS; evaluated horizon sensitivity, Kalman covariance, and counterfactual action choices.
- Reduced multiblock double-integrator total cost from 930.15 to 575.21 despite higher input cost, demonstrating the value of information-seeking actions.

Learning Belief Representations for Bilinear Observation Systems May 2026 – Oct 2026

- Built a structured autoregressive estimator of belief means and covariances from action-observation histories, without state or belief supervision or training-time Kalman filtering.
- Studied identifiability up to a fixed similarity transform; compared squared error with Gaussian negative log-likelihood to model uncertainty beyond point estimates.
- Achieved $R^2 = 0.9845$ on a six-state belief-mean task; learned-belief control recovered 64.7% of the oracle's 22.8% gain over certainty-equivalent control.

Cognitive Visual Reasoning Evaluation: The Prestige Mar 2026 – Jun 2026

McGill University / Cornell University | NeurIPS 2026

- Co-developed an open-ended benchmark with 2,177 questions from 180 magic videos to test temporal dependencies, object tracking, and event-level explanations.
- Evaluated 14 vision-language models against human baselines; analyzed missed events and object-identity errors as distinct sources of reasoning failure.
- On the same 200 questions, mean accuracy rose from 37.5% (open-ended) to 71.5% (four-choice) solely by changing response format, revealing sensitivity to the evaluation format.

SELECTED PROJECT

Crypto Liquidation Exchange Simulator Aug 2026 – Dec 2026

Cornell MFE Capstone with Coinbase Quantitative Risk Modeling

- Built a discrete-event order-book simulator from 193,781 SOL-USD top-of-book observations and a sampled 56.6-million-event BTC data set.
- Replayed 14,426 events without reconstruction inconsistencies; diagnosed 8,839 rejected orders, stale snapshots, and inventory-independent quotes as failure sources.
- Inventory-aware quote skew improved midpoint-latent-value correlation from 0.02 to 0.49 and reduced terminal inventory from 34.5 to 2.8 units; liquidation stress tests are ongoing.

HONORS

Dean's Honour List, McGill University 2022 – 2024

McGill Alumnae Susan Cameron Vaughan Scholarship ($2,000)

TECHNICAL SKILLS

Programming: Python, R, Java, C, MATLAB, Bash

Libraries: PyTorch, NumPy, pandas